

Aalok Tiwari

Quantitative data analysis & statistical modeling | Physics-aware machine learning workflows |
Experimental condensed matter physics (low dimensional topological devices)

EDUCATION

- 2023–present** **Carnegie Mellon University**, Pittsburgh, PA, USA
Ph.D. Candidate, Department of Physics (expected May 2028)
Advisor: Prof. Jyoti Katoch.
- 2018–2023** **Indian Institute of Science Education and Research**, Kolkata, India
Integrated BS–MS in Physics
Mentors: Prof. Goutam Dev Mukherjee (NCHPS) & Prof. Sayan Bhattacharyya (AFML).

EXPERTISE

Computing, data analysis & modeling: Developing physics-aware agentic AI workflows and statistical models leveraging machine learning algorithms to extract structure from large, multidimensional experimental datasets. Python, MATLAB, R, Igor Pro, Origin, GSAS, Gwyddion, Blender.

Synchrotron-based spectroscopy: XPS, ARPES, XAS, & X-PEEM

Lab-based characterization: UHV system operation, cryogenics, electrical transport, AFM, Raman and photoluminescence spectroscopy, XRD, FTIR, diamond anvil cell

2D device fabrication and cleanroom processes: heterostructure stacking, lithography, thin-film deposition (sputtering and evaporation)

Crystal growth: solid-state and solution-based reaction synthesis and composition control

EXPERIENCE

2023–present *Ph.D. Researcher, Katoch–Singh Lab, Carnegie Mellon University*

- ARPES and X-PEEM experimental design and execution at international light sources, in collaboration with multi-institutional research groups.
 - Successful beamtime allocation proposals.
 - ALS (USA), BNL (USA), BESSY II (Germany), Synchrotron SOLEIL (France) & ALBA (Spain).
- 2D device and heterostructure assembly via cleanroom fabrication and dry transfer stacking, including primary characterization and surface preparation ranging from inert to UHV environments.
- Presented research outputs at various conferences and workshops ([complete list](#)).

Many-body physics of field-effect devices

- Co-developing physics-aware agentic AI workflows and statistical models to streamline analysis of multi-dimensional ARPES data, turning raw multi-gigabyte detector output into quantitative band-structure results.
- Realized the first momentum-resolved electronic band structure of a functioning monolayer TaIrTe₄ device using μ ARPES.
- Collaboratively engineering *in situ* perturbation methods for real-time control of local strain and magnetic fields during ARPES measurements on bulk crystals to 2D devices.
- Band gap engineering of graphene in proximity to distinct substrates and their alignment for optoelectronic applications.

Imaging magnetic order at the nanoscale

- X-PEEM study of atomically thin and twisted CrSBr heterostructures.

- First element-specific visualization of layer parity dependent magnetism.
- Showed that spins in an orthogonally twisted bilayer undergo a magnetic reconstruction leading to an emergent magnetic order.
- Domain manipulation in an all-van der Waals spintronic device.
 - Real space map of current induced magnetization dynamics of a spintronic device.
 - Built a quantitative image-analysis pipeline to model and track the magnetization evolution.

2022–2023 Master's student, NCHPS & AFML Labs, IISER Kolkata

Low-temperature and high-pressure Raman studies of polycrystals

- Studied the anharmonic phonon contributions of cuprate superconductor YBCO in Raman and transport signatures across the superconducting transition (4–273 K).
- Synthesized vacancy-ordered double perovskite halides for solar cell applications and conducted high-pressure Raman and PL measurements using a diamond anvil cell (DAC).

AWARDS

MCS Graduate Conference Travel Award to attend the APS Global Physics Summit (2026).

Best Poster Award, Workshop on Scalable Analog Quantum Simulation Using Moiré Superlattices, organized by the Pittsburgh Quantum Institute (PQI) (2025).

PUBLICATIONS AND PREPRINTS

1. **Visualization of tunable electronic structure of monolayer TaIrTe₄**, S. A. Ekahana*, Tiwari, A.*, S. Sasmal*, et al., and J. Katoch, *Nature Communications* (accepted, 2026).
2. **Element-specific visualization of layer-parity and twist-dependent magnetism in CrSBr**, Tiwari, A.*, et al., S. Singh and J. Katoch, submitted for peer review (2026).
3. *[Review]* **Progress and prospect of artificial intelligence in ARPES workflow**, S. A. Ekahana*, Tiwari, A.*, et al., E. Rotenberg, and J. Katoch, to be submitted (2026).
4. **Current driven domain manipulation in all van der Waals spintronic device**, Tiwari, A., et al., S. Singh and J. Katoch, to be submitted (2026).
5. **In-plane anomalous Hall effect in a low-dimensional system**, I.-H. Kao*, R. K. Bandapelli*, et al., Tiwari, A., et al., J. Katoch and S. Singh, *Nature Materials* (2026).
6. **Tilted spin polarization enabled multidimensional unidirectional magnetoresistance**, R. K. Bandapelli*, I.-H. Kao*, et al., Tiwari, A., et al., and S. Singh, under review at *Physical Review Letters* (2026).
7. **Enabling electrical readout of Néel vector reversal in a van der Waals antiferromagnet**, R. Posti, R. K. Bandepalli, et al., Tiwari, A., et al., and S. Singh, submitted for peer review (2026).
8. **Imaging of van der Waals materials via standing-wave photoemission microscopy: depth-resolved electronic structure of WS₂**, J. R. Paudel, R. Muzzio, et al., Tiwari, A., et al., J. Katoch and A. X. Gray, under review at *Nano Letters* (2026).
9. **Structural control of defect-bound exciton emission in α -In₂Se₃**, R. Rao, P. Bhattacharya, et al., Tiwari, A., et al., and M. A. Susner, *Advanced Optical Materials* (accepted, 2026).
10. **Temperature- and pressure-dependent vibrational properties and phase stability of pristine and Sb-doped vacancy-ordered double perovskite**, Tiwari, A., K. Panigrahi, M. Sahu, S. Bhattacharyya, and G. D. Mukherjee, preprint on arXiv:2512.21810 (2025).
11. **Revealing the EuCd₂As₂ semiconducting band gap via n-type La-doping**, R. A. Nelson, J. King, S. Cheng, et al., Tiwari, A., et al., and J. E. Goldberger, *Chemistry of Materials* (2024). * denotes equal contribution.